

TraitHorizon: Scalable Exploration of Large Image-Feature Paired Datasets

Jackson Jacobs^{1*}, Fan Fan^{1*}, Laura Barisoni^{2,3}, and Andrew Janowczyk^{1,4,5}

¹ Department of Biomedical Engineering, Emory University and Georgia Institute of Technology, Atlanta, Georgia, USA ² Department of Pathology, Division of AI and Computational Pathology, Duke University, Durham, North Carolina, USA ³ Department of Medicine, Division of Nephrology, Duke University, Durham, North Carolina, USA ⁴ Division of Precision Oncology, Department of Oncology, University Hospital of Geneva, Geneva, Switzerland ⁵ Division of Clinical Pathology, Department of Diagnostics, University Hospital of Geneva, Geneva, Switzerland ¶ Corresponding author * These authors contributed equally.

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#)
- [Repository](#)
- [Archive](#)

Editor: [Open Journals](#)

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#))

Summary

Collections of Objects of Interest (OOIs)—such as cells, tubules, or tissue patches—paired with high-dimensional, computationally derived feature vectors are increasingly generated in image-based biomedical research. These OOI-feature datasets are routinely explored to detect outliers for quality control, uncover patterns, and generate biological insights. However, as datasets grow in size and complexity, researchers face bottlenecks: repeated manual creation of static visualizations for each subpopulation, difficulties efficiently plotting large numbers of high-dimensional feature vectors, and limited tooling for tracing individual feature values back to their source objects.

TraitHorizon is a browser-based visualization platform for interactive exploration of large OOI-feature pair datasets. It integrates three synchronized visualization components: (a) a parallel coordinates plot for feature-vector-level exploration, (b) dynamic violin plots for per-feature distribution analysis, and (c) a tabular data grid linking each feature vector to its corresponding object image. Its interactive interface enables real-time, scalable visualization of datasets containing hundreds of thousands of OOI-feature pairs. We demonstrate TraitHorizon's utility through a quality control case study involving 260,201 segmented kidney tubules, each characterized by 99 features, illustrating how the platform facilitates rapid, interpretable, and reproducible data interrogation.

TraitHorizon is publicly available at <https://traithorizon.com/>. The extended paper is available at https://github.com/choosehappy/TraitHorizon/blob/main/paper/paper_extended.pdf. TraitHorizon is available as a Python package on PyPI and as a Docker image, providing straightforward installation across computing environments.

Statement of need

Research in computational imaging fields often aims to discover trends in OOIs associated with diagnosis, prognosis, and therapy response (Y. Chen et al., 2023; Yijiang Chen et al., 2025; Fan Fan et al., 2025). From these objects, numerical feature vectors are extracted to encode their attributes, ranging from simple measures (e.g., area, texture) to deep learning-derived descriptors (Ambekar et al., 2025; Echle et al., 2021). These features are typically examined using static visualizations such as parallel coordinates plots (Heinrich & Weiskopf, 2013) or scatter plots of dimensionally-reduced object representations (e.g., using UMAP (McInnes

et al., 2020) or t-SNE (Maaten & Hinton, 2008)), aided by summary statistics, to uncover structure and outliers. Multiple subpopulations often emerge, prompting repeated visualization and metric generation.

This workflow remains time-intensive due to three limiting factors. First, there is a **lack of connection** between an object's image, feature values, and cohort-level visualization. Discovering outliers or clusters prompts the need to trace them back to images, but static plots force analysts to manually map plotted points to objects. Second, **exploring subpopulations incurs high time cost**: manually plotting each subpopulation is inefficient, while programmatic approaches require subpopulations to be defined *a priori*. Dynamic filtering and responsive plots are needed for iterative exploration. Third, **rendering latency disrupts analysis** when datasets reach hundreds of thousands of data points. Browser-based tools not designed for this scale suffer from heavy memory utilization, latency, and unresponsiveness.

These factors motivate the development of interactive visualization tools that dynamically generate visualizations during exploration, seamlessly link plotted data to source images, and efficiently render at scale.

State of the field

Several existing tools address parts of the interactive visualization problem but none fully satisfy all three requirements simultaneously. TensorBoard Projector (TensorFlow Team, 2025) and HoloViews (Installation — HoloViews V1.21.0, n.d.) support interactive plotting of large multivariate datasets but do not provide integrated image viewing linked to data points. DendroMap (Bertucci et al., 2022) facilitates qualitative exploration of large image collections but does not link images back to their feature vectors or support dynamic filtering by feature value. HistoQC (Janowczyk et al., 2019) provides image-linked plots and quality metrics but is tailored specifically to whole-slide-level quality control in digital pathology, supporting only its predefined feature set rather than arbitrary user-defined feature vectors or object-level images.

TraitHorizon was built as a standalone tool rather than contributing to these existing projects for several reasons. First, no existing tool couples image display with high-dimensional feature visualization and interactive filtering in a single interface. Second, TraitHorizon's input format—a simple TSV file paired with an image directory—is deliberately generic, making it compatible with any upstream tool that outputs tabular data (e.g., HistoQC (Janowczyk et al., 2019), CohortFinder (Fan et al., 2024)). Third, TraitHorizon's rendering pipeline was designed from the ground up for browser-based scalability to hundreds of thousands of objects, a requirement not addressed by the tools above.

Software design

TraitHorizon is a Flask-based web application (Pallets/Flask, n.d.), making it suitable for collaborative environments when hosted over a network. The front end leverages SlickGrid (SlickGrid Home, n.d.), Parallel Coordinates (parcoords (Yun, 2025)), and D3.js (D3 by Observable | The JavaScript Library for Bespoke Data Visualization, n.d.) to power the interactive dashboard. Its command-line interface ingests a directory of object images (any browser-supported format ("Image File Type and Format Guide - Media | MDN," 2025)) and a TSV file where each row contains an image filename followed by tab-separated feature columns. Integer, float, scientific notation, and categorical features are supported, along with an optional clickable URL column.

Key architectural decisions target scalability to datasets with hundreds of thousands of objects. (1) **Non-blocking progressive rendering** of the parallel coordinates plot keeps the interface responsive during rendering, completing in seconds to minutes depending on dataset size. (2) **Exportable filter configurations** allow parallel coordinate brush settings and filtered object IDs

to be saved and restored via a lean JSON format without re-rendering. (3) **Infinite scrolling** in the data grid dynamically loads and replaces content within a fixed memory footprint. (4) **On-demand image loading** limits browser memory and network utilization. (5) **On-demand violin plot computation** avoids unnecessary CPU usage by computing distribution plots only when a user hovers over a parallel axis.

These design choices enable TraitHorizon to operate smoothly within browser constraints on consumer-grade hardware, regardless of dataset size.

Use case

We demonstrate TraitHorizon using data from a digital pathology biomarker study (F. Fan et al., 2024; Fan Fan et al., 2025). The dataset comprises 260,201 segmented tubules and 99 associated pathomic features computed from 254 PAS-stained kidney biopsies from the NEPTUNE (Barisoni et al., 2013; Gadegbeku et al., 2013) and CureGN (CureGN Study Rationale, Design, and Methods, n.d.) consortia.

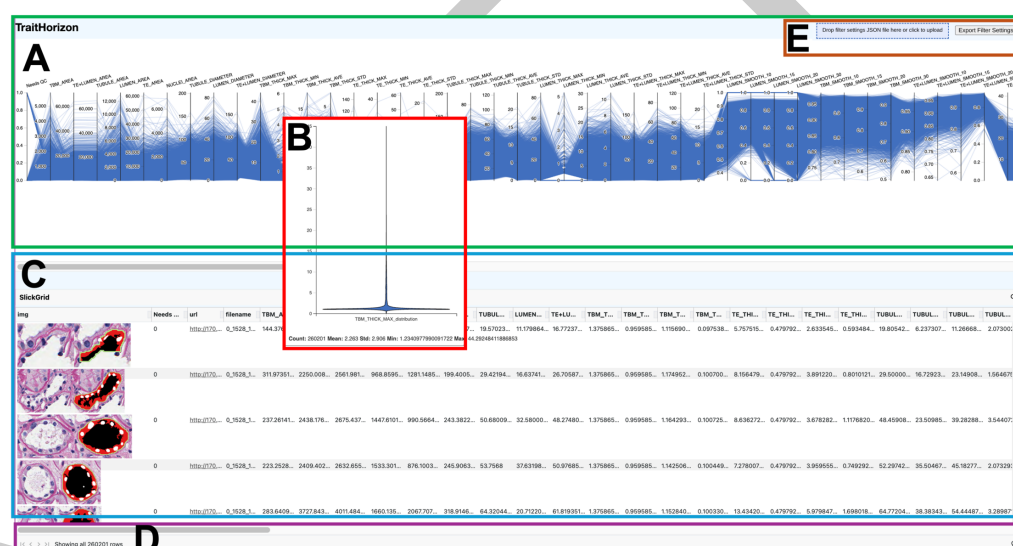


Figure 1: TraitHorizon web application interface. (A) The parallel coordinates plot displays 99 features from 260,201 tubules. (B) Violin plot of a selected feature on hover. (C) Data grid with segmented tubular images and associated features. (D) Status bar showing visible instance count. (E) Drop zone and export button for filter configurations.

After launching TraitHorizon (Figure 1), we explored the parallel coordinates plot for a global view of feature distributions. Hovering over TBM_THICK_MAX (maximum tubular basement membrane thickness) revealed a strongly skewed distribution via the violin plot, suggesting extremal-value tubules requiring closer examination.

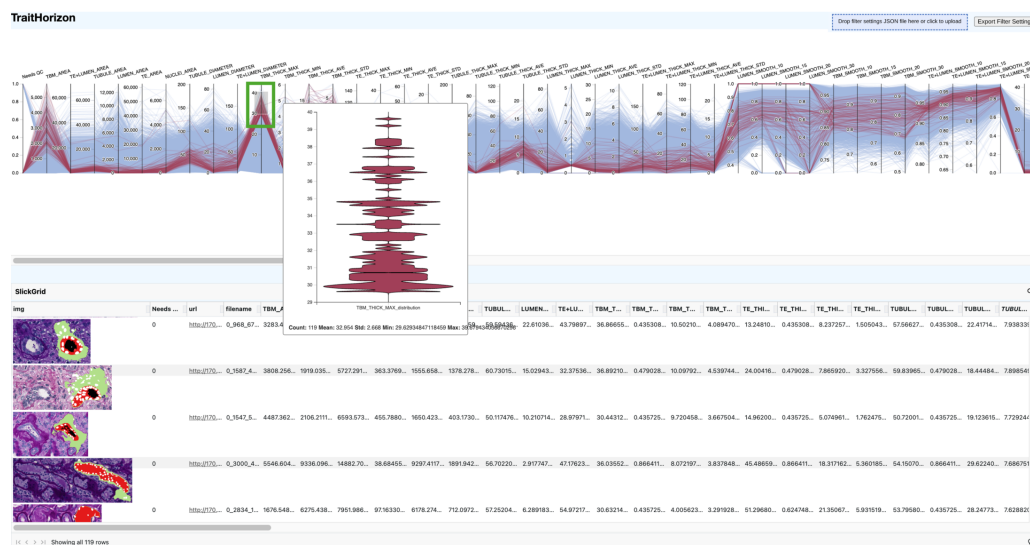


Figure 2: Interactive filtering isolating tubules with high TBM_THICK_MAX values (30–40). The 119 matching instances are highlighted in red; all others are greyed out.

We filtered tubules with high TBM_THICK_MAX by brushing the corresponding axis (Figure 2). The data grid revealed consistently thickened basement membranes (large green segments), a hallmark of tubular atrophy (*Automated Computational Detection of Interstitial Fibrosis, Tubular Atrophy, and Glomerulosclerosis | American Society of Nephrology*, n.d.; Fogo et al., 2016). Hovering over individual rows highlighted their feature vectors in the parallel coordinates plot (Figure 3), revealing a subpopulation exhibiting both thickened basement membranes and enlarged lumina. Applying a second brush on LUMEN_THICK_MAX (Figure 4) isolated this composite phenotype, which pathologists confirmed as a distinct morphological pattern (Fogo et al., 2016).

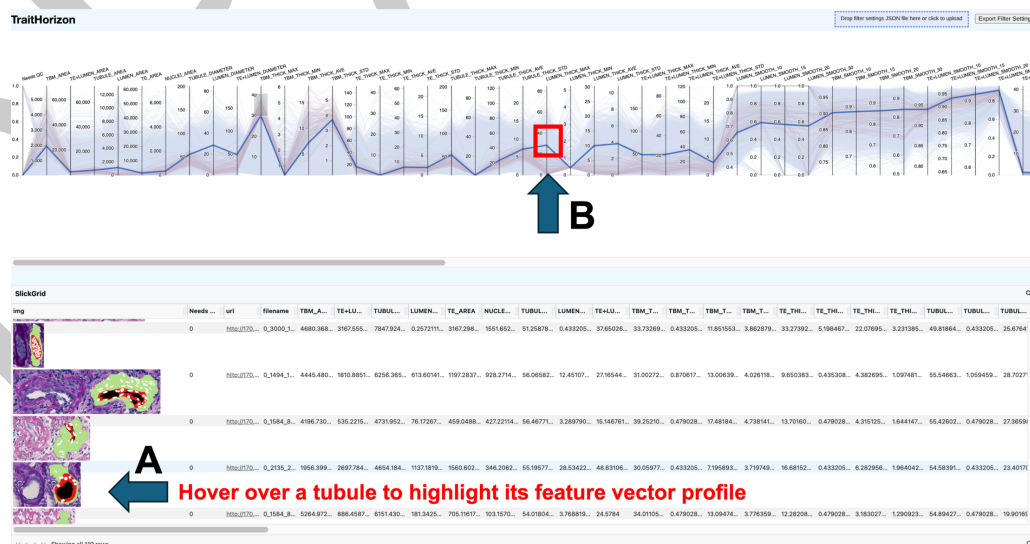


Figure 3: Hovering over an instance in the data grid highlights its feature vector in blue across the parallel coordinates plot, enabling multi-feature pattern identification.

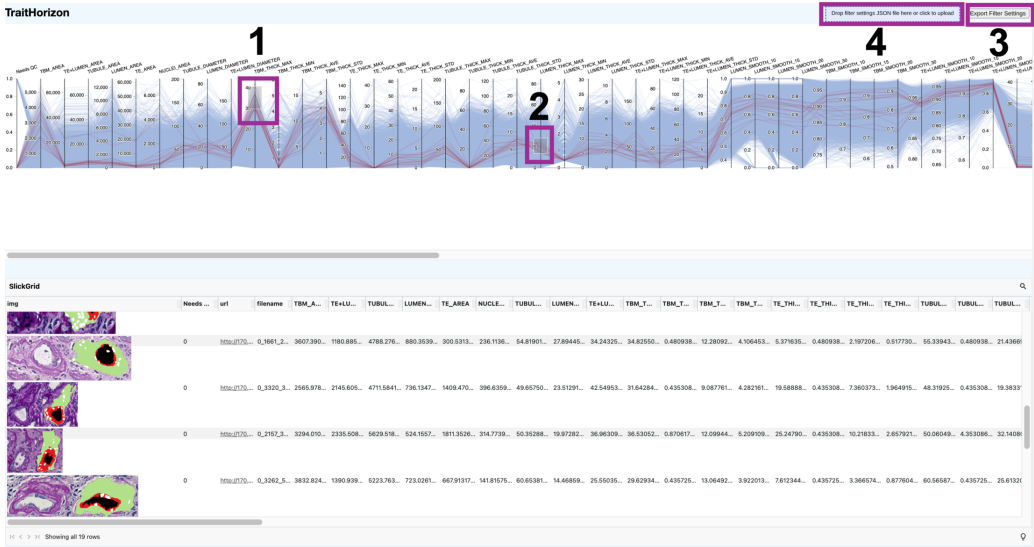


Figure 4: Joint filtering across TBM_THICK_MAX and LUMEN_THICK_MAX. Filter configurations can be exported (3) and reimported (4) as JSON for reproducibility.

115 TraitHorizon also supports exact-value search (Figure 6), sorting features within the data
116 grid (Figure 7), and linking objects to external viewers via URL columns (Figure 5).
117 In our workflow, clicking a URL opened the corresponding tubule in HistomicsUI
118 (*DigitalSlideArchive/HistomicsUI*, 2025) within the Digital Slide Archive (*Gutman et al.*,
119 2017), enabling pathologists to assess tubules at every scale from the immediate tissue
120 microenvironment to the whole slide context.

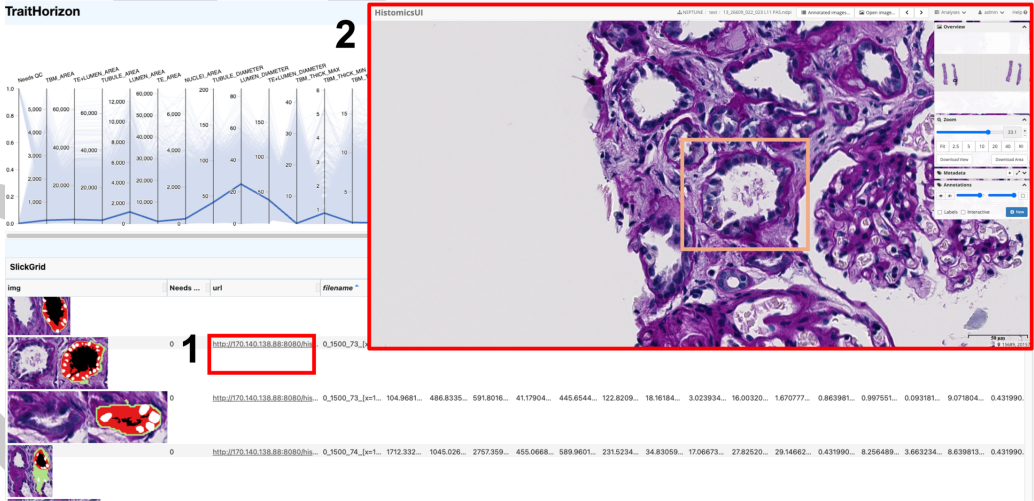


Figure 5: External URL integration linking a tubule to HistomicsUI for contextual visualization.

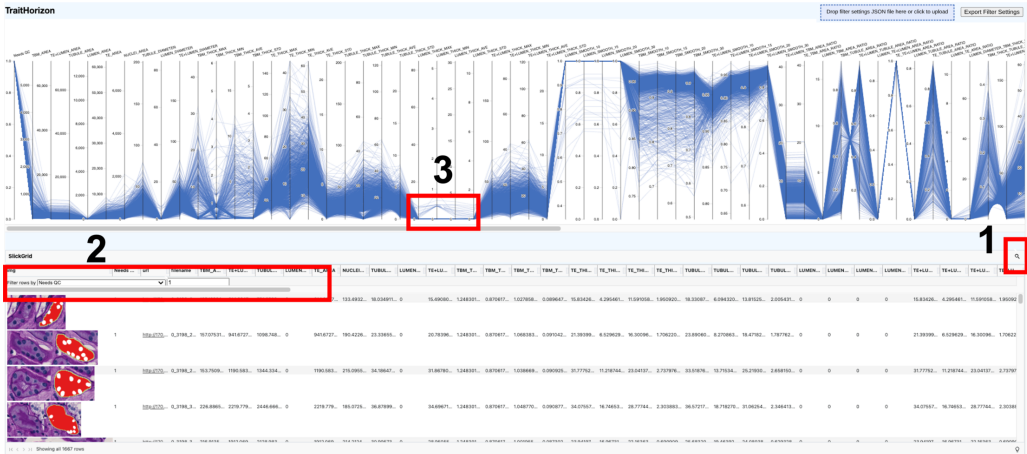


Figure 6: Searching by exact value using a “Needs QC” indicator to isolate flagged tubules.

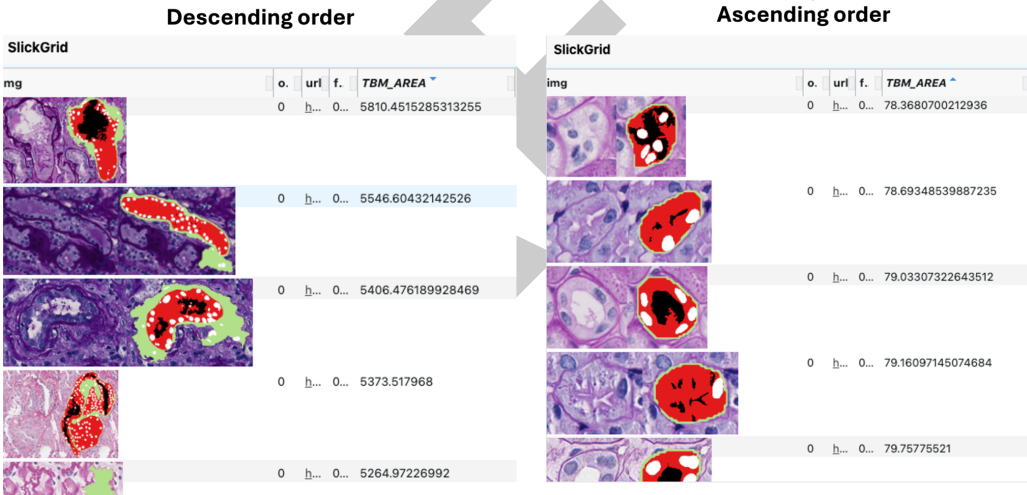


Figure 7: Sorting by TBM_AREA reveals a morphological gradient from thin to thickened basement membranes.

Research impact statement

TraitHorizon has already been leveraged in two published studies, where our collaborators investigated tubular pathomics in kidney disease. In Fan et al. (Fan et al., 2025), we used TraitHorizon to perform quality control and exploratory analysis of 260,201 segmented tubules and 99 pathomic features, uncovering morphological patterns along a trajectory from normal to atrophic tubules that were subsequently validated by study pathologists and found to be clinically relevant. In Ambekar et al. (Ambekar et al., 2025), we similarly used TraitHorizon for quality control of glomerular pathomic features in a study of minimal change disease and focal segmental glomerulosclerosis. In both studies, TraitHorizon enabled rapid identification of segmentation artifacts and biologically meaningful subpopulations that would have been impractical to discover using static visualization methods. The follow-up studies are as well taking advantage of its employ, as well as 7 others we are leading. A travel grant was awarded for the presentation of TraitHorizon at the annual Kidney Precision Medicine Project meeting, with the intent of broadening its dissemination.

AI usage disclosure

No generative AI tools were used in the development of the TraitHorizon software or for writing the original manuscript. Claude Opus 4 (Anthropic) was used solely to assist with restructuring and condensing the manuscript to meet the updated JOSS format and word limit requirements. Github Copilot was used to convert the TraitHorizon documentation from .rst format to markdown. All AI-generated text was reviewed, edited, and validated by the authors, who made all decisions regarding scientific content and framing.

Acknowledgements

Research reported in this publication was supported by:

- (1) the National Institutes of Health (NIH) under the following awards: R01LM013864 and R01DK118431;
- (2) The Nephrotic Syndrome Study Network (NEPTUNE) is part of the Rare Diseases Clinical Research Network (RDCRN), which is funded by the NIH and led by the National Center for Advancing Translational Sciences (NCATS) through its Division of Rare Diseases Research Innovation (DRDRI). NEPTUNE is funded under grant number U54DK083912 as a collaboration between NCATS and the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). Additional funding and/or programmatic support is provided by the University of Michigan, NephCure Kidney International, Alport Syndrome Foundation, and the Halpin Foundation. RDCRN consortia are supported by the RDCRN Data Management and Coordinating Center (DMCC), funded by NCATS and the National Institute of Neurological Disorders and Stroke (NINDS) under U2CTR002818;
- (3) Additional support was also provided by NephCure and the Henry E. Haller, Jr. Foundation;
- (4) Funding for the CureGN consortium is provided by U24DK100845, U01DK100846, U01DK100876, U01DK100866, and U01DK100867 from the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). Patient recruitment is supported by NephCure. Dates of funding for first phase of CureGN was 9/16/2013-5/31/2019. Dates of funding for the second phase of CureGN was 6/1/2019 - 5/31/2024. Date for the fundings for the third phase of CureGN are 6/1/2024-5/31/2029

References

- Ambekar, A., Roohian, M., Liu, Q., Wang, B., Fan, F., Cassol, C., Lafata, K., Holzman, L., Mariani, L., Hodgins, J., Zee, J., Janowczyk, A., & Barisoni, L. (2025). *Glomerular Segmentation, Classification, and Pathomic Feature-based Prediction of Clinical Outcomes in Minimal Change Disease and Focal Segmental Glomerulosclerosis*. medRxiv. <https://doi.org/10.1101/2025.10.01.25336172>
- Automated Computational Detection of Interstitial Fibrosis, Tubular Atrophy, and Glomerulosclerosis | American Society of Nephrology*. (n.d.). Retrieved August 29, 2022, from <https://jasn.asnjournals.org/content/32/4/837.abstract>
- Barisoni, L., Nast, C. C., Jennette, J. C., & others. (2013). Digital pathology evaluation in the multicenter Nephrotic Syndrome Study Network (NEPTUNE). *Clinical Journal of the American Society of Nephrology (CJASN)*, 8(8), 1449–1459. <https://doi.org/10.2215/CJN.08370812>
- Bertucci, D., Hamid, M. M., Anand, Y., Ruangrotsakun, A., Tabatabai, D., Perez, M., & Kahng, M. (2022). *DendroMap: Visual Exploration of Large-Scale Image Datasets for Machine Learning with Treemaps*. arXiv. <https://doi.org/10.48550/arXiv.2205.06935>

- 181 Chen, Yijiang, Wang, B., Demeke, D., Fan, F., Berthier, C., Mariani, L., Lafata, K., Holzman,
182 L., Hodgins, J., Janowczyk, A., Barisoni, L., & Madabhushi, A. (2025). Clinical relevance
183 of computational pathology analysis of interplay between kidney microvasculature and
184 interstitial microenvironment. *Clinical Journal of the American Society of Nephrology*,
185 20(2), 239–255. <https://doi.org/10.2215/CJN.0000000597>
- 186 Chen, Y., Zee, J., Janowczyk, A. R., Rubin, J., Toro, P., Lafata, K. J., Mariani, L. H.,
187 Holzman, L. B., Hodgins, J. B., Madabhushi, A., & Barisoni, L. (2023). Clinical relevance of
188 computationally derived attributes of peritubular capillaries from kidney biopsies. *Kidney360*,
189 4(5), 648–658. <https://doi.org/10.34067/KID.000000000000116>
- 190 *CureGN Study Rationale, Design, and Methods: Establishing a Large Prospective Observational*
191 *Study of Glomerular Disease - PubMed*. (n.d.). Retrieved May 30, 2024, from <https://pubmed.ncbi.nlm.nih.gov/30420158/>
192
- 193 *D3 by Observable | The JavaScript library for bespoke data visualization*. (n.d.). Retrieved
194 July 9, 2025, from <https://d3js.org/>
- 195 *DigitalSlideArchive/HistomicsUI*. (2025). Digital Slide Archive. [https://github.com/](https://github.com/DigitalSlideArchive/HistomicsUI)
196 [DigitalSlideArchive/HistomicsUI](https://github.com/DigitalSlideArchive/HistomicsUI)
- 197 Echle, A., Rindtorff, N. T., Brinker, T. J., Luedde, T., Pearson, A. T., & Kather, J. N. (2021).
198 Deep learning in cancer pathology: A new generation of clinical biomarkers. *Br J Cancer*,
199 124(4), 686–696. <https://doi.org/10.1038/s41416-020-01122-x>
- 200 Fan, F., Liu, Q., Zee, J., & others. (2024). Computationally derived characterization of tubular
201 changes in relation to the development of interstitial fibrosis: FR-PO933. *Journal of the*
202 *American Society of Nephrology*, 35(10S). <https://doi.org/10.1681/ASN.20248v5y8bwn>
- 203 Fan, Fan, Liu, Q., Zee, J., Ozeki, T., Demeke, D., Yang, Y., Bitzer, M., O'Connor, C. L.,
204 Farris, A. B., Wang, B., Shah, M., Jacobs, J., Mariani, L., Lafata, K., Rubin, J., Chen, Y.,
205 Holzman, L., Hodgins, J. B., Madabhushi, A., ... Janowczyk, A. (2025). Clinical relevance of
206 computationally derived tubular features and their spatial relationships with the interstitial
207 microenvironment in minimal change disease/focal segmental glomerulosclerosis. *Kidney*
208 *International*, 0(0). <https://doi.org/10.1016/j.kint.2025.04.026>
- 209 Fan, Fan, Martinez, G., DeSilvio, T., Shin, J., Chen, Y., Jacobs, J., Wang, B., Ozeki, T.,
210 Lafarge, M. W., Koelzer, V. H., Barisoni, L., Madabhushi, A., Viswanath, S. E., &
211 Janowczyk, A. (2024). CohortFinder: An open-source tool for data-driven partitioning
212 of digital pathology and imaging cohorts to yield robust machine-learning models. *Npj*
213 *Imaging*, 2(1), 15. <https://doi.org/10.1038/s44303-024-00018-2>
- 214 Fogo, A. B., Lusco, M. A., Najafian, B., & Alpers, C. E. (2016). AJKD atlas of renal
215 pathology: Ischemic acute tubular injury. *American Journal of Kidney Diseases*, 67(5), e25.
216 <https://doi.org/10.1053/j.ajkd.2016.03.003>
- 217 Gadegbeku, C. A., Gipson, D. S., Holzman, L. B., Ojo, A. O., Song, P. X. K., Barisoni, L.,
218 Sampson, M. G., Kopp, J. B., Lemley, K. V., Nelson, P. J., Lienczewski, C. C., Adler, S.
219 G., Appel, G. B., Cattran, D. C., Choi, M. J., Contreras, G., Dell, K. M., Fervenza, F. C.,
220 Gibson, K. L., ... Kretzler, M. (2013). Design of the Nephrotic Syndrome Study Network
221 (NEPTUNE) to evaluate primary glomerular nephropathy by a multidisciplinary approach.
222 *Kidney Int*, 83(4), 749–756. <https://doi.org/10.1038/ki.2012.428>
- 223 Gutman, D. A., Khalilia, M., Lee, S., & others. (2017). The digital slide archive: A software
224 platform for management, integration, and analysis of histology for cancer research. *Cancer*
225 *Research*, 77(21), e75–e78. <https://doi.org/10.1158/0008-5472.CAN-17-0629>
- 226 Heinrich, J., & Weiskopf, D. (2013). State of the art of parallel coordinates. In M. Sbert &
227 L. Szirmay-Kalos (Eds.), *Eurographics 2013 - state of the art reports*. The Eurographics
228 Association. <https://doi.org/10.2312/conf/EG2013/stars/095-116>

- 229 Image file type and format guide - Media | MDN. (2025). In *MDN Web Docs*. [https://](https://developer.mozilla.org/en-US/docs/Web/Media/Guides/Formats/Image_types)
230 developer.mozilla.org/en-US/docs/Web/Media/Guides/Formats/Image_types
- 231 *Installation — HoloViews v1.21.0*. (n.d.). Retrieved July 3, 2025, from <https://holoviews.org/>
- 232 Janowczyk, A., Zuo, R., Gilmore, H., Feldman, M., & Madabhushi, A. (2019). HistoQC: An
233 Open-Source Quality Control Tool for Digital Pathology Slides. *JCO Clin Cancer Inform*,
234 3, CCI.18.00157. <https://doi.org/10.1200/CCI.18.00157>
- 235 Maaten, L. van der, & Hinton, G. (2008). Visualizing Data using t-SNE. *Journal of Machine*
236 *Learning Research*, 9(86), 2579–2605. <http://jmlr.org/papers/v9/vandermaaten08a.html>
- 237 McInnes, L., Healy, J., & Melville, J. (2020). *UMAP: Uniform Manifold Approximation and*
238 *Projection for Dimension Reduction*. arXiv. <https://doi.org/10.48550/arXiv.1802.03426>
- 239 *Pallets/flask: The Python micro framework for building web applications*. (n.d.). Retrieved
240 October 28, 2025, from <https://github.com/pallets/flask/?tab=readme-ov-file>
- 241 *SlickGrid Home*. (n.d.). Retrieved July 9, 2025, from <https://slickgrid.net/Index.html>
- 242 TensorFlow Team. (2025). *Visualizing data using the embedding projector in TensorBoard*.
243 https://www.tensorflow.org/tensorboard/tensorboard_projector_plugin.
- 244 Yun, X. (2025). *BigFatDog/parcoords-es*. <https://github.com/BigFatDog/parcoords-es>

DRAFT